

Watching a 3.8-billion-parameter transformer think: a browser-native, reference-validated, 1:1 real-time visualization of an LLM forward pass

Ahmet Barış Günaydın *Independent researcher · github.com/abgnydn/neuropulse · neuropulse.live*

Keywords: WebGPU; WGSL; large language models; transformer; mechanistic interpretability; in-browser computation; visualization; reproducibility; pre-registration; attention fixed points

Abstract

Interactive visualizations of transformer language models fall into two disjoint regimes: pedagogical tools that animate *toy* models — on the order of 10^3 parameters, a sort-the-alphabet task — in faithful 3D, and inference engines that run real multi-billion-parameter models with *no* internal visibility. The two have never met: one can watch a model that is not real, or run a real model one cannot watch. **neuropulse** closes that gap. It renders a real forward pass of Phi-3-mini-4k-instruct (3.8 B parameters, 32 layers) in a web browser, in real time, with every rendered element bound one-to-one to a named tensor in the model’s compute graph and read back from the same WebGPU buffers the model just wrote — no interpolation, no sampling, no schematic stand-in, and no server. To our knowledge this is the first visualization that is simultaneously (i) a real multi-billion-parameter model, (ii) in-browser, (iii) in 3D, (iv) over *all* live tensors rather than attention alone, and (v) validated against a reference implementation. The engine is hand-written: eleven WGSL compute kernels over MLC’s q4f16_1 weights, 292 GPU dispatches per token headless and 348 with the visualizer instrumented. The one-to-one claim is made falsifiable: a built-in suite diffs the live WebGPU forward pass against a pinned HuggingFace fp16 reference at nine layer checkpoints and holds per-layer relative $L_2 < 2 \times 10^{-2}$, cosine > 0.999 , **100 % top-1 token agreement**, and softmax Jensen–Shannon divergence $< 5 \times 10^{-3}$ — the reader’s own GPU is the test rig. We treat the platform as an empirical lab with a pre-registered prediction log and demonstrate the payoff with experiment E45, a falsification of a continuous-attention self-consistency hypothesis: iterating Phi-3-mini’s attention to its Picard fixed point destroys language coherence at a sharp cliff (iteration 2, universal across 16/16 prompts) and converges to a degenerate low-dimensional attractor, whereas a well-trained 4-layer control model retains 91 % of its behavior — locating the brittleness in model scale and training breadth rather than in transformers as such.

1. Introduction

Most public “AI visualizations” are decoration: particle systems and pulsing dots with no model behind them, showing how a designer *imagines* a language model works. A smaller set of serious tools show something real, but each surrenders a dimension that matters for understanding a *production-scale* model. Bycroft’s LLM-Viz renders a transformer in 3D with live tensors — of a $\sim 10^3$ -parameter toy that sorts letters. Transformer Explainer [1] runs a real GPT-2-small (124 M) live in the browser, in 2D, with partial tensor exposure. BertViz [2] surfaces real attention from real HuggingFace models, but attention only, in a notebook, in 2D. In-browser inference runtimes such as WebLLM run real multi-billion-parameter models at full speed and expose nothing inside.

The gap is structural. Visualization tools scale *down* to stay legible; inference engines treat internals as opaque to stay fast. neuropulse takes the position that the internals of a *full-scale* model, drawn faithfully, justify the engineering required to expose them — as pedagogy (one can watch a real 3.8 B transformer generate a token) and as an interpretability substrate (every tensor is addressable, so experiments run *on the artifact being shown*, not on a proxy).

Contributions. This work contributes:

1. **A browser-native, server-free engine** that runs a real Phi-3-mini forward pass as eleven hand-written WGSL kernels over 4-bit weights, with a derived (not hard-coded) dispatch budget of 292 per token (§2.1).
2. **A one-to-one rendering architecture** in which the renderer reads the inference buffers directly rather than recomputing, so on-screen geometry *is* the model state; spatial layout is derived from the model’s own weights (§2.2).
3. **A falsifiable validation methodology** — a per-layer diff against a pinned HuggingFace fp16 reference that the reader runs on their own GPU, plus a commit-time gate that fails the build when documentation drifts from source (§2.3, §3.2).
4. **An empirical-lab demonstration (experiment E45)** that uses the platform to pre-register and falsify a continuous-attention self-consistency hypothesis, and — via a trained small-model control — attributes the resulting brittleness to scale and training breadth (§3.3).

The contribution is neither a new model nor a new rendering primitive; it is the demonstration that a real, full-scale forward pass can be rendered one-to-one in a browser tab, that the claim can be made falsifiable against a reference, and that the resulting apparatus is a usable interpretability lab.

2. Methods

2.1 Inference engine

The model is Phi-3-mini-4k-instruct [3]: 3.8 B parameters; 32 decoder layers; 32 attention heads (no grouped-query attention); head dimension 96; hidden dimension 3,072; SwiGLU feed-forward inner dimension 8,192; vocabulary 32,064. It is loaded in MLC’s q4f16_1 ndarray-cache format — 4-bit weights, fp16 scales, group size 32 — *not* GGUF. The decode loop is hand-written as eleven WGSL compute kernels (embedding, int4_matmul, int4_matmul_f32, rope, attention, attention_scores, kv_append, rms_norm, add_norm, fused_ffn, argmax) with no inference framework in the path. One generated token costs

$$D_{\text{fast}} = 9L + 4 = 292, \quad D_{\text{viz}} = D_{\text{fast}} + L + 8 \cdot 3 = 348 \quad (L = 32),$$

where the headless cost is nine per-layer dispatches plus a four-dispatch prologue/epilogue, and the visualized path adds one attention-scores readback per layer and eight three-dispatch logit-lens probes. Counts are derived in source from a closed form, not written as prose (§2.3). Weights stream once and persist in OPFS and the Cache API for instant reload.

2.2 One-to-one rendering by shared-buffer readback

Inference and visualization **share the same GPU buffers** (Figure 1). The renderer does not recompute, approximate, or re-simulate: it reads the values the model already produced and maps each to a scene element bound to a named tensor — the residual stream, the 32×32 attention-head grid, the gated MLP, the token strip, the per-layer logit-lens. Spatial layout is derived from the model rather than chosen by a designer: residual-stream positions are the two-dimensional PCA projection of the model’s own layer-0 qkv_proj weight matrix, so dimensions read into attention together cluster together. Ten draggable panels each surface one live tensor (top- k , confidence, KV-cache occupancy, residual norm, per-layer delta, last-layer attention, logit-lens, and so on), persisted to localStorage.

2.3 Validation methodology

“Strict one-to-one” is a strong claim, so it is made falsifiable. A built-in suite diffs the live WebGPU forward pass against a pinned HuggingFace fp16 Phi-3-mini at nine checkpoints, $\mathcal{V} = \{0, 4, 8, 12, 16, 20, 24, 28, 31\}$ (27,648 floats compared per prompt). For a layer ℓ with live hidden state h_{ℓ}^{GPU} and reference h_{ℓ}^{HF} over the leading 3,072 dims, the suite asserts

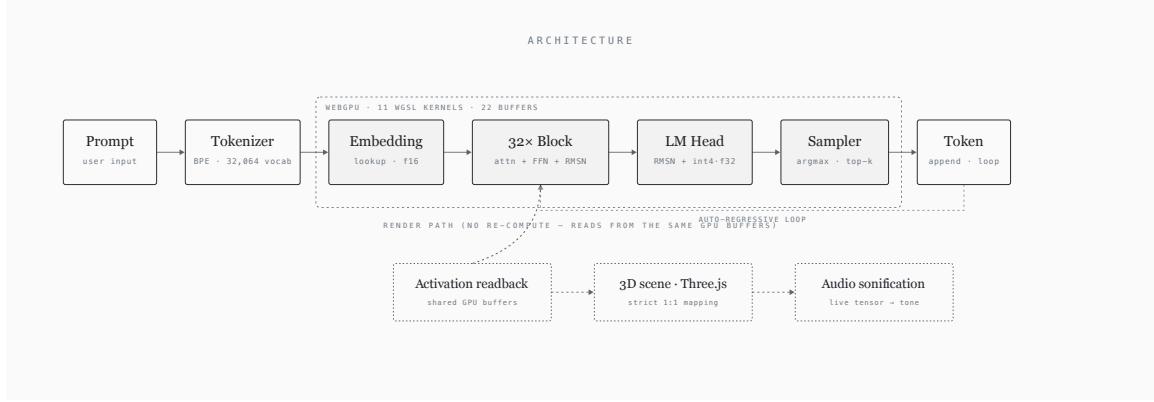


Figure 1: System architecture. The auto-regressive inference path (prompt, tokenizer, embedding, 32 transformer blocks, LM head, sampler) and the render path read the *same* GPU buffers; the renderer surfaces the activations the model already produced rather than recomputing them.

$$\frac{\|h_{\ell}^{\text{GPU}} - h_{\ell}^{\text{HF}}\|_2}{\|h_{\ell}^{\text{HF}}\|_2} < 2 \times 10^{-2}, \quad \cos(h_{\ell}^{\text{GPU}}, h_{\ell}^{\text{HF}}) > 0.999.$$

The single intentional precision tradeoff versus the FP32 reference is f16 accumulation of intermediate residuals at every Add+Norm boundary; matmul, softmax, RMSNorm ($\epsilon = 10^{-5}$), RoPE, and SiLU each accumulate at f32 inside the kernels. A second, commit-time gate cross-checks every numeric claim in the documentation (layer count, kernel count, dispatch counts, keyboard shortcuts) against source, failing the build on drift, so prose cannot silently diverge from code.

2.4 Experimental protocol

All measurements are on an Apple M2 Pro under Chromium with WebGPU. Decoding is deterministic greedy (argmax) for every parity and E45 measurement; sampling is exercised only by the RNG self-test. Each run is stamped with a runtime fingerprint — build SHA, GPU vendor/architecture, and browser version — surfaced in the demo footer and recorded with every reported artifact, so any number can be regenerated under a known configuration. The reference dump is produced by `tools/dump_phi3_reference.py` against a pinned HuggingFace revision; the parity suite runs from the demo’s validate button in under a minute on the reader’s own GPU. E45 (§3.3) uses the existing 15-prompt logit sweep plus a 16-prompt multi-format sweep (English, code, mathematics, Japanese, emoji, JSON), 12 tokens per prompt, run through a headed-Chromium Playwright harness; the small-model control runs on CPU.

3. Results

3.1 The visualization system

neuropulse renders a real Phi-3-mini forward pass at interactive rates in a browser tab: a prompt is typed, and 3.8 B parameters process it with every intermediate tensor read back and drawn. Five view modes (Journey, Scene, Attention, Logit-Lens, Cinematic) and four stacking overlays present the same underlying activations from different angles; a keyboard-driven cinematic flythrough, camera control, and the panel set cover the interaction surface. Because the render path reads the inference buffers directly (§2.2), what is on screen *is* the model’s state rather than a depiction of it. Table 1 positions the system against prior interactive transformer visualizations.

Table 1: Interactive transformer visualizations across six axes. neuropulse is, to our knowledge, the first to satisfy all six simultaneously.

Tool	Real model	Scale	Browser	3D	Live tensors	Ref-validated
LLM-Viz (Bycroft)	toy (sorts ABC)	$\sim 10^3$	yes	yes	yes	no
Transformer Explainer [1]	GPT-2-small	124 M	yes	no	partial	no
BertViz [2]	HF models	any	no	no	attention only	no
WebLLM	yes	multi-B	yes	—	no	—
neuropulse	Phi-3-mini	3.8 B	yes	yes	all	HF fp16

3.2 Correctness against a reference

The one-to-one claim holds against HuggingFace fp16 within the declared tolerances (Table 2); a run exceeding any bound is flagged a regression. The hidden-state deltas are the cost of int4 quantization and f16 residual round-trips, not implementation drift; the decisive bar — identical top-1 tokens versus the fp16 reference on the validation prompt set — is met. Each kernel additionally carries a declared per-kernel error budget (e.g. $\text{int4_matmul} \leq 4$ ULP at f16, $\text{rope} \leq 1$ ULP, kv_append and embedding bit-exact); wiring the per-kernel CPU references into continuous integration is a tracked gap (§4).

Table 2: Validation bounds asserted by the in-app suite against a pinned HuggingFace fp16 Phi-3-mini reference.

Test	Quantity	Bound
Hidden state vs HF	relative L_2 per layer (3,072 dims)	$< 2 \times 10^{-2}$ (obs. max $\approx 1.4 \times 10^{-2}$)
Hidden state vs HF	cosine similarity per layer	> 0.999
Last-attention reconstruction	rel. L_2 of $(\sum \text{scores} \cdot V - \text{attn_out})$	$< 1 \times 10^{-2}$
Top-1 token agreement	greedy argmax, ≤ 30 tokens	100 % match
Top-5 overlap	mean shared top-5 tokens (GPU vs HF)	≥ 4
Softmax divergence	mean JSD over top- k	$< 5 \times 10^{-3}$
Sampler self-test	JSD(5,000 samples, softmax)	$< 1 \times 10^{-2}$

3.3 What the lab produces: experiment E45

Because the platform exposes every tensor and ships a pre-registered prediction log, hypotheses can be tested on the model being visualized. Prediction P-20260526-07 asked whether iterating Phi-3-mini’s attention to a per-layer self-consistency (Picard) fixed point of

$$Q \leftarrow \text{softmax}(QK^\top / \sqrt{d}) V,$$

with RoPE applied once at iteration 0 and K, V held fixed, yields hidden states close to the standard one-step operator. The fixed-point kernel ships behind a flag (`?attn=fixedpoint`); a wiring gate first confirmed that `max_iter = 1` reproduces the baseline byte-for-byte, before any claim was made.

The hypothesis is **falsified at zero-shot inference**. One additional Picard step destroys coherence at a sharp cliff (Figure 2). On “The capital of Japan is ...”, iteration 1 matches the standard kernel exactly (“...Tokyo. Tokyo is a major global”), iteration 2 collapses (“lee\n\n\nTop \nTop”), and iterations 3–100 converge byte-for-byte to a degenerate topic-projector attractor. A 16-prompt multi-format sweep makes the cliff universal: **16/16 prompts coherent at iteration 1 collapse at iteration 2**, with a low-dimensional attractor vocabulary (“lee”, “ício”, “RESS”, newlines) and a second failure mode (immediate stop-token / empty output) on two prompts. The fixed point demonstrably *exists* and is numerically reachable — the max-norm change between successive iterates falls to f32 noise within 100 iterations — at only $1.41\times$ the compute of one step, so the cliff is the phenomenon, not an artifact of under-iteration or a compute budget.

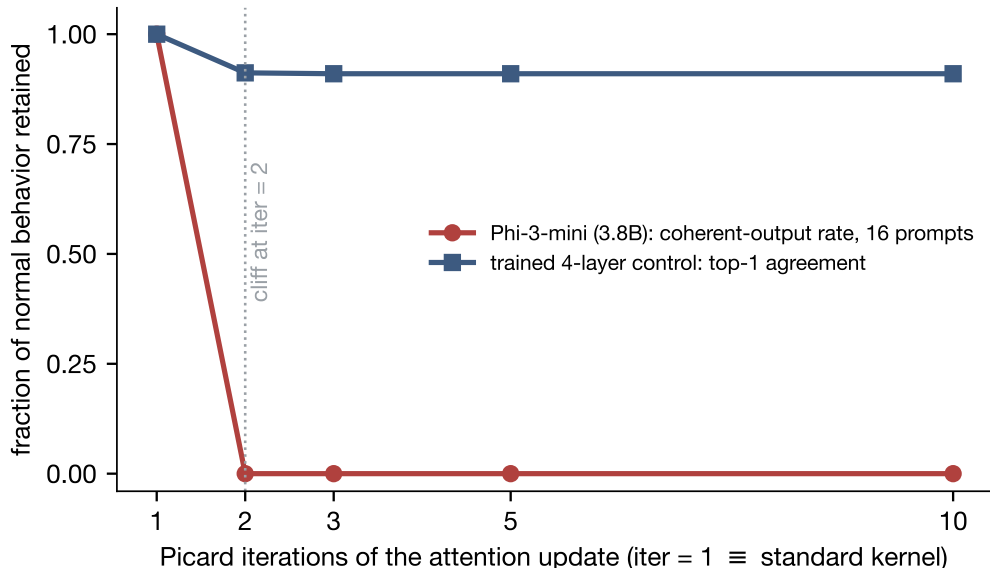


Figure 2: The Picard cliff. Iterating Phi-3-mini’s attention past one step destroys language coherence at iteration 2 (all 16 prompts collapse), while a well-trained 4-layer control under the identical protocol retains 91 % top-1 agreement and is stable through 100 iterations. Iteration 1 equals the standard kernel by construction. The two series use different but comparable “behavior retained” metrics (see text); the qualitative gap is the result.

A disentanglement control separates scale from the phenomenon. A well-trained 4-layer / 64-hidden toy transformer (same RMSNorm + SwiGLU + RoPE family) under the identical protocol stays in the “match” regime — 91 % top-1 retained at iteration 2, stable through iteration 100 — where Phi-3-mini retains no coherent output (Figure 2). The defensible claim is therefore stronger than “the hypothesis is false”: *a trained discrete-attention layer’s distance from its own Picard fixed point grows with model scale and training breadth; brittleness to attention iteration is a property of well-trained large LMs specifically, not of transformers in general*. This relates the inference-time behavior of trained attention to the equilibrium view of attention as a Hopfield update [4] and to deep equilibrium models [5], and shows the trained operator is qualitatively unlike its own one-step fixed point. A per-layer single-layer-ablation (Phase 3) to adjudicate the two pre-registered per-layer hypotheses is the natural follow-up and is not yet run.

4. Limitations and threats to validity

- **Per-kernel error budgets are declared, not yet CI-asserted.** The bounds of §3.2 are documented and the end-to-end HF parity is enforced on every user click, but the runner that checks each kernel against a CPU f64 reference does not yet exist; the per-kernel ULP figures are therefore design budgets, not measurements.
- **Cross-vendor parity is under-sampled.** The suite has been exercised on Apple M-series and one

NVIDIA workstation. WebGPU f16 behavior can differ across adapters; the cross-vendor matrix is open, and users are asked to report fingerprint plus parity numbers. This is the main external-validity threat.

- **One architecture.** The engine is specialized to Phi-3-mini and the MLC q4f16_1 format, whose RoPE base/theta-scaling it inherits; generalization to other architectures is unimplemented.
- **E45 metric heterogeneity.** Figure 2 compares a coherent-output rate (Phi-3, 16 prompts) against top-1 agreement (control); the two are not the same statistic. They are both “fraction of normal behavior retained,” and the qualitative gap (0 % vs 91 %) is far larger than any reasonable metric reconciliation, but a single shared metric across scales is future work.
- **E45 per-layer attribution is deferred.** The output cliff at iteration 2 is sharp enough that the two pre-registered per-layer hypotheses (deepest-middle vs high-entropy layers as catastrophic) are both partially refuted at model-level granularity; the single-layer ablation that would pick a winner is Phase 3, unrun.
- **Not a training or safety tool.** No claim is made about the model’s fairness, alignment, or safety — only that what is rendered matches the reference forward pass within the stated bounds.

5. Related work

Prior interactive transformer visualizations trade scale for visibility or vice versa (Table 1): LLM-Viz (real 3D, live tensors, $\sim 10^3$ -parameter toy), Transformer Explainer [1] (real GPT-2-small, 124 M, 2D, partial tensors), BertViz [2] (real attention only, 2D, notebook), and in-browser runtimes such as WebLLM (real multi-billion-parameter models, no internal visibility). Across the axes *real model* · *multi-billion scale* · *browser* · *3D* · *all live tensors* · *reference-validated*, neuropulse is, to our knowledge, the first to satisfy all six simultaneously. The interpretability result of §3.3 connects to equilibrium/fixed-point views of attention: modern Hopfield networks identify the attention update with a Hopfield retrieval step whose fixed points are energy minima [4], and deep equilibrium models train networks *defined* by a fixed-point solve [5]. We instead probe the *inference-time* self-consistency point of an already-trained discrete-attention model and find it qualitatively unlike the one-step operator, with a severity that grows with scale.

6. Reproducibility and software availability

Source: github.com/abgnydn/neuropulse (MIT); live at neuropulse.live. The WGSL kernels are in `src/engine/shaders/`; canonical architecture constants and the derived dispatch formulas in `src/engine/compiler.ts` and `src/engine/phi3-facts.ts`; the precision matrix, tolerances, and per-kernel budgets in `METHODS.md`; the pre-registered prediction log in `PREDICTIONS.md`. The HF parity suite runs from the demo’s validate button on any WebGPU-capable Chromium and regenerates its reference via `tools/dump_phi3_reference.py`; the E45 artifacts (iteration sweep, multi-prompt sweep, small-model control) are committed under `tests/results/` and `tools/small-model-control/`. Every documentation number is gated against source by `npm run verify` on each commit. The Butterfly context-compaction study that also ran on this platform has its own repository (github.com/abgnydn/butterfly) and deposit.

Data availability. All experimental artifacts are committed to the repository; the figures in this manuscript are reproducible from those artifacts. Model weights are Microsoft’s Phi-3-mini redistributed by MLC and are not part of this deposit.

Author contributions. A.B.G. is the sole author and conducted all design, implementation, experiments, and writing.

Generative-AI disclosure. Portions of the software, documentation, and this manuscript were drafted with a large language model used as a coding and writing aid; all output was author-reviewed, correctness was enforced by the HF parity suite and the commit-time claim checks, and every quantitative claim is traceable to `METHODS.md`, `phi3-facts.ts`, or a committed experiment artifact.

Statements. Sole author; no competing interests; no external funding.

7. Conclusion

A full-scale transformer forward pass can be rendered one-to-one, in real time, in a browser tab — every element a real tensor read from the buffers the model just wrote, every “one-to-one” claim falsifiable against a HuggingFace reference the reader runs on their own GPU. Closing the gap between toy-but-visible and real-but-opaque yields both a pedagogical artifact and an interpretability substrate: experiment E45 used it to falsify a continuous-attention self-consistency hypothesis and, via a trained small-model control, to attribute the resulting brittleness to scale and training breadth rather than to transformers in general. The platform’s discipline — derived constants, a pre-registered prediction log, and a validation suite that fails loud — is as much the contribution as the rendering.

References

- [1] A. Cho, G. Kim, A. Karpekov, A. Helbling, Z. J. Wang, S. Lee, B. Hoover, and D. H. Chau. “Transformer Explainer: Interactive Learning of Text-Generative Models.” *arXiv:2408.04619*, 2024. <https://arxiv.org/abs/2408.04619>
 - [2] J. Vig. “A Multiscale Visualization of Attention in the Transformer Model.” *Proc. 57th Annual Meeting of the ACL: System Demonstrations*, 2019, pp. 37–42. *arXiv:1906.05714*. <https://aclanthology.org/P19-3007/>
 - [3] M. Abdin et al. (Microsoft). “Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone.” *arXiv:2404.14219*, 2024. <https://arxiv.org/abs/2404.14219>
 - [4] H. Ramsauer, B. Schäfl, J. Lehner, P. Seidl, M. Widrich, T. Adler, L. Gruber, M. Holzleitner, M. Pavlović, G. K. Sandve, V. Greiff, D. Kreil, M. Kopp, G. Klambauer, J. Brandstetter, and S. Hochreiter. “Hopfield Networks is All You Need.” *Int. Conf. on Learning Representations (ICLR)*, 2021. *arXiv:2008.02217*. <https://arxiv.org/abs/2008.02217>
 - [5] S. Bai, J. Z. Kolter, and V. Koltun. “Deep Equilibrium Models.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. *arXiv:1909.01377*. <https://arxiv.org/abs/1909.01377>
 - [6] W3C GPU for the Web Working Group. “WebGPU.” W3C Candidate Recommendation. <https://www.w3.org/TR/webgpu/>
-

Draft v0.2. Built from the committed engine, METHODS.md tolerances, and the PREDICTIONS.md / tests/results/ E45 artifacts; every documentation number is gated against source by npm run verify. To be converted to a venue LaTeX template before submission to a visualization (IEEE VIS / EuroVis), ML-systems, or interpretability venue.